

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Artificial Intelligence**

Code No: **CSC266** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define admissible heuristic with an example. Explain the working mechanism and limitations of hill climbing search.
2. How do you define problem? What are criteria for defining problem? Compare constraint Satisfaction Problem and Real-World Problem in detail with appropriate example.
3. Define Expert System with example. What are stages of expert system development? Explain.

Section B

Attempt any Eight questions[8×5=40]

4. How syntactic and semantic analyses are performed in natural language processing?
5. What do you mean by Rational Agent? What are differences between Utility based agent and model-based agent?
6. What is state space representation? Illustrate with one example.
7. What is forward chaining? Explain with appropriate example.
8. Convert following sentences into predicatea. All animals who can bark are dog.b. Someone is firing a gunc. All tigers are not fierce
9. Define game. Write the benefits and limitations of depth limited search.
10. What is fuzzy logic? Discuss the different operators used in genetic algorithm.
11. Give an example of reinforcement learning. Explain the types of ANN.
12. Write short note of the following (Any TWO):a. Pragmatic Analysisb. Unification and liftingc. Turing test